

[Collaborative Docs](#) >

Discussion Agenda 01/13, 02/09, 03/09/2021 (CosmoRun+CGM)

Below is an ongoing discussion agenda / live progress report for AGORA Paper Groups "CosmoRun+CGM" and others for monthly web conferences.

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Recent important discussions that require your attention are in red color!

(1st revision: 02/09/2021)

[1. Paper Group Progress Report \(as of Feb. 2021\)](#)

[\(1\) Ongoing Projects](#)

	Paper "CosmoRun" + Paper "CGM"	Paper "HighresRun" + Paper "Clumps"																																								
GOALS	• Paper "CosmoRun" : Introduce ~1e12 Ms halo run, emphasis on the calibration process and how it differs from previous attempts like Aquila, discuss basic properties of halos • Paper "CGM" : Study absorption lines, focus on CGM properties that are not resolution-dependent (e.g., high ions)	• Paper "HighresRun" : Run ~1e12 Ms halo at higher resolution (20 pc instead of 80 pc) to study SF clumps at high z, nice resolution study when compared to Paper "CosmoRun" • Paper "Clumps" : Study star-forming clumps, compare with observations																																								
LEADERS	• Cameron Hummels (Caltech) • Santi Roca-Fabrega (UCM)	• Daniel Ceverino (Heidelberg) • Alessandro Lupi (IAP)																																								
PARTICIPANTS	• Code representatives: <table><tr><th>Code</th><th>Contacts</th></tr><tr><td>ART-I</td><td>Dekel, Mandelker, Primack, Roca-Fabrega, Strawn</td></tr><tr><td>ART-II</td><td>(Feldmann)</td></tr><tr><td>CHANGA</td><td>Powell, Quinn</td></tr><tr><td>ENZO</td><td>Barrow, Hummels, Kim (Ji-hoon), Kim (Ki-won), O'Shea, Shin, Wise</td></tr><tr><td>GADGET</td><td>Nagamine, Shimizu</td></tr><tr><td>GASOLINE</td><td>(Madau, Shen, Wadsley, Wissing)</td></tr><tr><td>GEAR</td><td>Hausammann, Revaz</td></tr><tr><td>GIZMO</td><td>Dong, Hopkins, Hummels, Lupi, Velazquez</td></tr><tr><td>RAMSES</td><td>Agertz, Roca-Fabrega, Teyssier</td></tr></table> (Code leaders as of Dec. 2020) - ICs help: Buehlmann, Hahn, Simpson - Analysis help: Chakrabarti, Fumagalli, Hummels, Kim, Madau, Nagai, Roca-Fabrega, Smith, Turk, Strawn	Code	Contacts	ART-I	Dekel, Mandelker, Primack, Roca-Fabrega, Strawn	ART-II	(Feldmann)	CHANGA	Powell, Quinn	ENZO	Barrow, Hummels, Kim (Ji-hoon), Kim (Ki-won), O'Shea, Shin, Wise	GADGET	Nagamine, Shimizu	GASOLINE	(Madau, Shen, Wadsley, Wissing)	GEAR	Hausammann, Revaz	GIZMO	Dong, Hopkins, Hummels, Lupi, Velazquez	RAMSES	Agertz, Roca-Fabrega, Teyssier	• Code representatives: <table><tr><th>Code</th><th>Contacts</th></tr><tr><td>ART-I</td><td>Ceverino, Dekel, Lee, Mandelker, Primack, Roca-Fabrega</td></tr><tr><td>ART-II</td><td>(Feldmann)</td></tr><tr><td>CHANGA</td><td>(Quinn)</td></tr><tr><td>ENZO</td><td>Kim (Ji-hoon), O'Shea</td></tr><tr><td>GADGET</td><td>Nagamine, Shimizu</td></tr><tr><td>GASOLINE</td><td>(Mayer, Wadsley)</td></tr><tr><td>GEAR</td><td>Hausammann, Revaz</td></tr><tr><td>GIZMO</td><td>Dong, Hopkins, Lupi</td></tr><tr><td>RAMSES</td><td>Agertz, Roca-Fabrega</td></tr></table> (Code leaders as of Nov. 2019) - ICs help: Buehlmann, Hahn - Analysis help: Chakrabarti, Kim, Turk, Roca-Fabrega, Strawn • Working Groups engaged: - III (common cosmo ICs) - IV (common analysis) - XI (high-z galaxies)	Code	Contacts	ART-I	Ceverino, Dekel, Lee, Mandelker, Primack, Roca-Fabrega	ART-II	(Feldmann)	CHANGA	(Quinn)	ENZO	Kim (Ji-hoon), O'Shea	GADGET	Nagamine, Shimizu	GASOLINE	(Mayer, Wadsley)	GEAR	Hausammann, Revaz	GIZMO	Dong, Hopkins, Lupi	RAMSES	Agertz, Roca-Fabrega
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	• Working Groups engaged: – III (common cosmo ICs) – IV (common analysis) – V (subgrid physics) – IX (characteristics of cosmo. galaxies) – X (outflows)	
PROGRESS MADE OR BEING MADE	• Visit Paper "CosmoRun"+"CGM" Workspace for the latest updates including issues being faced by the group and the individual code groups, and the Overleaf draft being written. • Code leaders have submitted datasets for the proposed cosmological run aimed at calibrating the feedback at $z=4$ first. This includes various calibration datasets. • Paper Group "CosmoRun"+"CGM" has had ~30 meetings/telecons in the past ~3 years! If you haven't, you may talk to Santi or Ji-hoon for the recent analysis results, how to contribute to this effort, etc. Any code representative who is not participating in the current "CosmoRun" comparison is especially welcomed!	• Visit Paper "HighresRun"+"Clumps" Workspace for the latest updates. • For the Papers in the stage after "CosmoRun" or "CGM", -- such as Papers "HighresRun", "Clumps", and more -- see their relevant Workspaces, or talk to Ji-hoon Kim with any question.
OTHER LINKS	• Paper Workspace • Trident download and documentation • Trident method paper (Hummels et al. 2016)	• Paper Workspace

2. More Items to Discuss

(1) Workshop This Year (And Next Year) (2020–2021)

- We have come to an agreement that we will hold our next AGORA Workshop when the draft of Paper "CosmoRun" is ready and when the scopes and strategies of the next Papers can be discussed. The workshop's format will very likely be online, and it will be announced on this mailing list in a timely fashion.
- With the completion of Paper "CosmoRun", we aim to relaunch/reinitiate the 9 Science-oriented Working Groups (science topics first identified in 2012–2013; more information in [this link](#)). Due to various changes in the past years, it is expected that the Science-oriented Working Group leadership positions will be reopened for younger postdocs and students to grab in the next year's Workshop.
- We also expect to have a much larger workshop in 2021(?) with wider science topics (i.e., not like the technically very focused ones in the past two years). So please stay tuned. And if possible, start contributing your time and work to Paper "CosmoRun" and Paper "CGM".

(2) AGORA Resources on NERSC:

- We thank the annual effort by Primack and Abel to secure our continuous access to NERSC computing resources. Currently our allocation for the year is 200 TB storage and half a million cpu hours (for analysis job only!). We also concluded that applying for simulation times at NERSC is probably unnecessary.

(3) Future Science-oriented Paper Ideas

- (3-A) Offshoot Paper Ideas Out of Paper "CosmoRun" (80 pc resolution):
 - Morphology: comparison with observation (using e.g., Huertas-Company's tool)
 - Emission lines: focus on a few lines that observers will be interested in (using e.g., Barrow's tool which he is willing to share and possibly merge into yt)
- (3-B) Offshoot Paper Ideas Out of Paper "HighresRun" (20 pc resolution):
 - Dark matter: previously Science-oriented Working Group VII (pitched in this workshop by Jiang/Freundlich; a DM-only run needed along with the 20 pc resolution "HighresRun"; see also "Core/Cusp" in [this link](#))
 - Satellite galaxies: previously Science-oriented Working Group VIII
 - Dwarf galaxies: previously Science-oriented Working Group VI (pitched in this workshop by Hausammann; a new run needed with a 10^{10} Ms IC)
- (3-C) Offshoot Paper Ideas Out of Our Paper "Disk" (Isolated Disk Run Paper Previously Published):
 - See Paper "Disk-II" in the table down below (Section 4-(8)).
 - See also "Secular Evolution" in [this link](#).

3. Recaps

(1) How to Request an Account on NERSC:

- If you represent a code group but don't have a NERSC user account, please visit [this page](#). Select "standard" account and find the repository name "agora". Joel Primack will add your name to the group when approved.

(2) Strategy to Store Simulation Outputs (A Default Policy; Strategies May Vary Per Project):

- Please visit [this link](#).

(3) AGORA SMBH Prescription for Accretion and Feedback:

- Please visit [this link](#).

(4) Your Own yt Installation on NERSC:

- Please visit [this link](#).

(5) yt-Sunrise Pipeline for AGORA on NERSC / Working Installation of Sunrise as Docker Image:

- Please visit [this link](#).

(6) Mini-workshop Opportunities:

- Mini-workshop for individual paper groups to write papers: KIPAC@Stanford is willing to provide funding opportunities for working group leaders to organize mini-workshops (~10 people). An example was the [yt-AGORA mini-workshop](#) at UCSC in Mar. 2014.

(7) Other Useful Links:

- Quick links to the important pages in AGORA Workspace, including internal policies: [here](#)
- Internal policy regarding the disclosure of AGORA simulation results: [here](#)
- Collection of softwares relevant to AGORA effort: [here](#)

(8) Previously Launched Projects:

Paper "Disk-II"		Paper "Metallicity"																														
GOALS	• Compare simulation codes using common disk ICs	• Run ~1e12 Ms halo to study metal distribution																														
LEADERS	• Santi Roca-Fabrega (UCM)																															
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PROGRESS MADE OR BEING MADE	• Visit Paper's Workspace where the results for disk comparison are compiled (including those presented in our first disk comparison paper – previously known as Paper "4"). The agora-analysis-script repository has been accordingly updated.	• Visit Paper's Workspace for the latest updates.																														
	• Analysis items that were not covered in our first disk comparison paper (see this link) will be considered in the next paper, Paper "Disk-II".	• Will piggyback on the planned Paper "CGM" runs and evolve them further down to z=0.																														
	• A subset of topics could be discussed in the "Metal Diffusion" or "Secular evolution" paper (see this link).																															
OTHER LINKS	• Progress report in Paper's Workspace and/or in WG V's Workspace . • yt-AGORA Google group	• Paper Workspace																														

	Paper "DM-Highres"	Paper "NoSubgrid"
GOALS	• Run Proof-of-concept-DM (~1e11 Ms halo) at	• Run Proof-of-concept-NOSUBGRID (~1e11

	higher res	Ms halo)																																														
LEADERS	• Oliver Hahn (OCA, Nice) • Christine Simpson (HITS)	• Nick Gnedin (Fermilab) • Piero Madau (UCSC) • Britton Smith (SDSC)																																														
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PROGRESS MADE OR BEING MADE	• Visit Paper's Workspace where Simpson, Hahn and others have been updating the progress. • We will plan to have 5-code comparison. As of Aug. 2017, 4+ participating groups have completed and submitted the production-level results (~100 pc resolution) to NERSC. • We may focus on the (survival of) substructures. This subset of topics could potentially to be led by Jiang as a smaller paper. • Consistent tree – python interface has been built by Britton Smith (ytree).	• Code groups such as ART-I and GIZMO have carried out cosmological simulations with various AGORA ICs, even including sophisticated subgrid physics.																																														
OTHER LINKS	• Progress report in Paper's Workspace and/or in WG III's Workspace. • MUSIC Google group • ytree documentation	• GRACKLE Google group • GRACKLE download & documentation • GRACKLE method paper (Smith et al. 2017) • AGORA API for GRACKLE by Nick Gnedin																																														

[5. Your Feedback](#)

Let us hear anything AGORA, including the scientific and organizational effort currently being made. Voice your opinion during the web conference! Feel free to share your thoughts in the comments section below, or on the [mailing list](#). You can also directly contact the [project coordinator](#) anytime!